Predicting Future Dementia Conversion from Baseline Clinical and MRI-Derived Biomarkers Using Interpretable Machine Learning

Independent Computational Neuroscience Research Project

Author: Simran Bhatnagar

Abstract

Background: Early identification of individuals at risk for dementia is an important challenge in neuroscience and clinical medicine. Machine learning has emerged as a promising approach for integrating demographic, cognitive, and neuroimaging data to predict disease progression.

Objective: This study investigated whether baseline demographic characteristics, cognitive assessments, and MRI-derived structural biomarkers could predict future dementia conversion using the OASIS-2 longitudinal dataset.

Methods: A retrospective longitudinal analysis was performed using baseline data from 86 participants (72 stable nondemented individuals and 14 future converters). Demographic variables, Mini-Mental State Examination (MMSE), Clinical Dementia Rating (CDR), and MRI-derived biomarkers (estimated total intracranial volume [eTIV], normalized whole-brain volume [nWBV], and Atlas Scaling Factor [ASF]) were analyzed. Statistical comparisons included Welch's t-tests, Cohen's d, and Pearson correlation analyses. Logistic Regression and Random Forest models were evaluated using repeated stratified five-fold cross-validation.

Results: No individual baseline variable demonstrated statistically significant differences between stable participants and future converters. Logistic Regression consistently outperformed Random Forest, achieving a mean ROC-AUC of 0.556 while providing greater model interpretability. Correlation and feature importance analyses indicated that dementia conversion is influenced by the combined contribution of multiple baseline variables rather than a single dominant predictor.

Conclusions: Baseline demographic, cognitive, and MRI-derived measurements provided modest predictive information regarding future dementia conversion. Although predictive performance was limited by the relatively small cohort size, the findings demonstrate the value of combining traditional statistical analyses with interpretable machine learning to investigate neurodegenerative disease progression. Future studies using larger cohorts and multimodal biomarkers may improve predictive performance and clinical applicability.

Keywords: Alzheimer's disease; dementia; machine learning; Logistic Regression; Random Forest; MRI; OASIS-2; neurodegeneration; predictive modeling.

1. Introduction

Dementia is one of the leading causes of disability and loss of independence among older adults worldwide. It is characterized by progressive deterioration in memory, reasoning, language, and other cognitive functions, ultimately interfering with an individual's ability to perform everyday activities. Alzheimer's disease is the most common cause of dementia, accounting for approximately 60–80% of all dementia cases (Alzheimer's Association, 2025). As populations continue to age, the number of individuals living with dementia is expected to increase substantially, creating significant medical, social, and economic challenges.

One of the greatest challenges in treating dementia is that the underlying neurodegenerative processes often begin many years before noticeable clinical symptoms appear. By the time a patient receives a diagnosis, substantial neuronal loss and structural brain changes may have already occurred. Consequently, researchers have increasingly focused on identifying biomarkers capable of detecting disease progression during its earliest stages, when therapeutic interventions may be more effective.

Structural magnetic resonance imaging (MRI) has become one of the most widely used techniques for studying neurodegenerative diseases (Frisoni et al., 2010) because it provides detailed measurements of brain anatomy without exposing patients to ionizing radiation. MRI-derived biomarkers—including measures of brain volume and structural atrophy—have been associated with cognitive decline and disease progression in numerous studies. In addition to neuroimaging, clinical assessments such as the Mini-Mental State Examination (MMSE) and Clinical Dementia Rating (CDR) provide standardized measures of cognitive and functional impairment that are routinely used in both research and clinical practice (Folstein et al., 1975; Morris, 1993).

Recent advances in machine learning have created new opportunities to combine demographic information, cognitive assessments, and neuroimaging biomarkers to improve prediction of future dementia progression (Vieira et al., 2017). Unlike traditional statistical approaches that often evaluate variables independently, machine learning algorithms can integrate multiple sources of information simultaneously to identify patterns associated with disease risk. However, developing reliable predictive models remains challenging because many publicly available longitudinal datasets are relatively small and contain limited numbers of participants who later develop dementia.

The objective of this study was to investigate whether baseline demographic characteristics, cognitive assessments, and MRI-derived structural biomarkers could predict future dementia conversion using the OASIS-2 longitudinal dataset. Baseline measurements were analyzed using both traditional statistical methods and supervised machine learning approaches. Logistic Regression and Random Forest models were evaluated and compared using repeated stratified cross-validation to determine whether interpretable machine learning could identify meaningful predictors of future dementia conversion.

2. Literature Review

Dementia and Alzheimer's Disease

Dementia is a syndrome characterized by progressive impairment in memory, reasoning, language, and other cognitive functions that significantly interfere with daily life. Alzheimer's disease is the most common cause of dementia and accounts for approximately two-thirds of all diagnosed cases (Alzheimer's Association, 2025). As global populations continue to age, the prevalence of dementia is expected to increase substantially, making early diagnosis and intervention an important priority for neuroscience research and public health.

The pathological processes underlying Alzheimer's disease begin many years before clinical diagnosis and include the accumulation of amyloid-β plaques, neurofibrillary tangles composed of hyperphosphorylated tau protein, synaptic dysfunction, and progressive neuronal loss (DeTure & Dickson, 2019). These biological changes gradually produce structural brain atrophy and cognitive decline that become increasingly detectable as the disease progresses.

MRI Biomarkers in Neurodegeneration

Structural magnetic resonance imaging (MRI) has become one of the most widely used tools for investigating neurodegenerative disease because it provides detailed, non-invasive measurements of brain anatomy. Previous research has demonstrated that reductions in hippocampal volume, cortical thinning, ventricular enlargement, and overall brain atrophy are associated with cognitive decline and progression toward dementia (Frisoni et al., 2010).

Although regional MRI biomarkers often demonstrate stronger predictive performance than global measures, whole-brain measurements such as normalized whole-brain volume (nWBV) and estimated total intracranial volume (eTIV) remain valuable because they are readily available in many public datasets and reflect broad structural changes associated with aging and neurodegeneration.

Machine Learning for Dementia Prediction

Over the past decade, machine learning has become an increasingly important approach for predicting cognitive decline and dementia progression. Algorithms including Logistic Regression, Support Vector Machines, Random Forests, Gradient Boosting, and deep neural networks have been applied to demographic information, cognitive assessments, neuroimaging data, and molecular biomarkers (Vieira et al., 2017).

Despite promising results, many published studies rely on relatively large datasets or incorporate specialized biomarkers such as hippocampal segmentation, amyloid PET imaging, cerebrospinal fluid biomarkers, or genetic information. These resources may not always be available in routine clinical settings, motivating continued investigation of models that rely on more commonly collected clinical and MRI-derived measurements.

Interpretable Machine Learning

While increasingly complex machine learning models often achieve high predictive accuracy, they may provide limited insight into how predictions are generated. In clinical research, interpretability is particularly important because physicians must understand the factors influencing model predictions before incorporating them into decision-making.

Logistic Regression remains one of the most widely used interpretable machine learning algorithms because model coefficients can be directly translated into odds ratios that quantify the relationship between predictors and clinical outcomes (Hosmer et al., 2013). This interpretability makes Logistic Regression particularly attractive for biomedical applications involving relatively small datasets.

Research Gap

Previous studies have demonstrated that combining clinical assessments with neuroimaging biomarkers can improve prediction of cognitive decline. However, relatively few investigations have examined the predictive value of readily available baseline demographic characteristics, cognitive assessments, and global MRI-derived measurements within a small longitudinal cohort while emphasizing model interpretability.

The present study addresses this gap by evaluating whether baseline clinical and MRI-derived variables can predict future dementia conversion using interpretable machine learning methods. Rather than prioritizing predictive performance alone, this study also examines feature importance and model interpretability to better understand which baseline characteristics contribute to future dementia risk.

3. Methods

Study Design

This study employed a retrospective longitudinal observational design to investigate whether baseline demographic characteristics, cognitive assessments, and MRI-derived structural biomarkers could predict future dementia conversion. Rather than diagnosing existing dementia, the objective was to determine whether information collected during a participant's initial clinical evaluation contained sufficient predictive information to identify individuals who would later develop dementia.

The analytical workflow consisted of four major stages. First, the dataset was cleaned and a prediction cohort was constructed using only baseline participant visits. Second, descriptive statistics and statistical hypothesis testing were performed to compare stable participants with individuals who later converted to dementia. Third, machine learning models were developed using baseline demographic, cognitive, and MRI-derived variables. Finally, model performance and interpretability were evaluated using multiple statistical and machine learning metrics.

Dataset

This study utilized the Open Access Series of Imaging Studies (OASIS-2) longitudinal MRI dataset, a publicly available dataset containing repeated structural magnetic resonance imaging (MRI) examinations and accompanying clinical assessments from older adults with and without dementia (Marcus et al., 2010). The dataset includes demographic variables, cognitive assessments, and global MRI-derived structural measurements collected across multiple clinical visits.

Because the objective of this study was to predict future dementia conversion using only information available at baseline, only the first recorded visit for each participant was included in the prediction analysis. Restricting the analysis to baseline measurements ensured that all predictor variables would have been available prior to disease progression and prevented information from future visits from influencing the prediction models.

Cohort Selection

The original OASIS-2 dataset contained longitudinal MRI sessions from 150 unique participants. Because multiple visits were available for many participants, records were first sorted chronologically according to participant identification number and visit number. The earliest visit for each participant was then selected as the baseline observation.

Participants who were already classified as demented at baseline were excluded from the prediction cohort because the objective of the study was to predict future dementia conversion rather than classify existing disease. The final prediction cohort therefore consisted of participants who remained nondemented throughout follow-up and participants who were classified as "Converted," indicating that they were nondemented at baseline but later developed dementia.

The final cohort included 86 participants, consisting of 72 stable nondemented individuals and 14 future converters.

Variables

Baseline predictor variables included demographic characteristics, cognitive assessments, and MRI-derived structural measurements. Demographic variables consisted of participant age, biological sex, years of education (EDUC), and socioeconomic status (SES). Cognitive performance was assessed using the Mini-Mental State Examination (MMSE), while functional impairment was evaluated using the Clinical Dementia Rating (CDR).

Structural MRI-derived variables included estimated total intracranial volume (eTIV), normalized whole-brain volume (nWBV), and Atlas Scaling Factor (ASF). Together, these variables provided complementary information regarding brain structure, cognitive function, and participant demographics.

The outcome variable was future dementia conversion. Participants classified as "Converted" within the OASIS-2 dataset were assigned a value of 1, while participants who remained nondemented throughout follow-up were assigned a value of 0.

Data Preprocessing

Prior to analysis, the dataset was inspected for missing values, variable types, and duplicate participant records. Because the prediction task required only baseline information, the earliest visit for each participant was retained while subsequent visits were excluded.

Continuous predictor variables were standardized before machine learning using z-score normalization within the preprocessing pipeline. Missing numerical values were imputed using the median value of the corresponding variable, while categorical variables were imputed using the most frequently observed category. Categorical variables were encoded using one-hot encoding to permit inclusion within the machine learning models.

Embedding preprocessing within the machine learning pipeline ensured that all transformations were performed independently within each cross-validation fold, thereby preventing data leakage.

Statistical Analysis

Baseline differences between stable participants and future converters were evaluated using Welch's independent-samples t-tests. Welch's t-test was selected because it does not assume equal variances between comparison groups and is generally more robust for datasets with unequal sample sizes.

To complement hypothesis testing, Cohen's d was calculated for each continuous variable to quantify the magnitude of group differences independently of sample size. Pearson correlation coefficients were subsequently calculated to examine relationships among baseline variables and their associations with future dementia conversion.

Statistical significance was defined as $p < 0.05$ for all analyses.

Machine Learning

Two supervised machine learning algorithms were evaluated: Logistic Regression (Hosmer et al., 2013) and Random Forest (Breiman, 2001). Logistic Regression was selected because of its interpretability and widespread use in clinical prediction studies, while Random Forest was included as a nonlinear ensemble learning method capable of modeling complex interactions among predictor variables.

To minimize overfitting and ensure reproducibility, model performance was evaluated using repeated stratified five-fold cross-validation. Stratification preserved the proportion of stable participants and future converters within each fold, while repeated cross-validation provided more stable estimates of model performance than a single train-test split.

Model performance was assessed using receiver operating characteristic area under the curve (ROC-AUC), balanced accuracy, precision, recall, and F1-score. Logistic Regression coefficients, odds ratios, and permutation feature importance were examined to improve model interpretability and identify the variables contributing most strongly to prediction.

Software

All analyses were conducted using Python 3 within the Google Colab environment. Data manipulation was performed using pandas and NumPy, statistical analyses were conducted using SciPy, machine learning models were implemented using scikit-learn, and data visualizations were generated using Matplotlib.

4. Results

Cohort Characteristics

Following cohort selection, 86 participants met the inclusion criteria for analysis. The final prediction cohort consisted of 72 participants who remained nondemented throughout follow-up and 14 participants who were classified as future converters. Future converters represented approximately 16.3% of the final cohort.

Baseline demographic characteristics, cognitive assessments, and MRI-derived structural measurements are summarized in Table 1.

Table 1. Baseline demographic, cognitive, and MRI-derived characteristics of stable participants and future converters.

Variable	Stable mean	Converted mean	t-statistic	p-value	Cohen’s d	Significant
Age	75.431	77.071	-0.722	0.479	0.201	False
MMSE	29.194	29.357	-0.608	0.551	0.189	False
CDR	0.0	0.036	-1.000	0.336	0.679	False
eTIV	1480.111	1438.286	1.005	0.325	-0.237	False
nWBV	0.746	0.738	0.840	0.411	-0.221	False
ASF	1.203	1.229	-0.775	0.446	0.187	False

Baseline Statistical Comparisons

Welch's independent-samples t-tests were performed to compare baseline variables between stable participants and future converters. No variable demonstrated a statistically significant difference between groups ($p > 0.05$ for all comparisons).

Although none of the observed differences reached statistical significance, effect size analysis suggested that baseline Clinical Dementia Rating (CDR) demonstrated the largest standardized difference between groups, while age and MRI-derived measurements exhibited relatively small effect sizes.

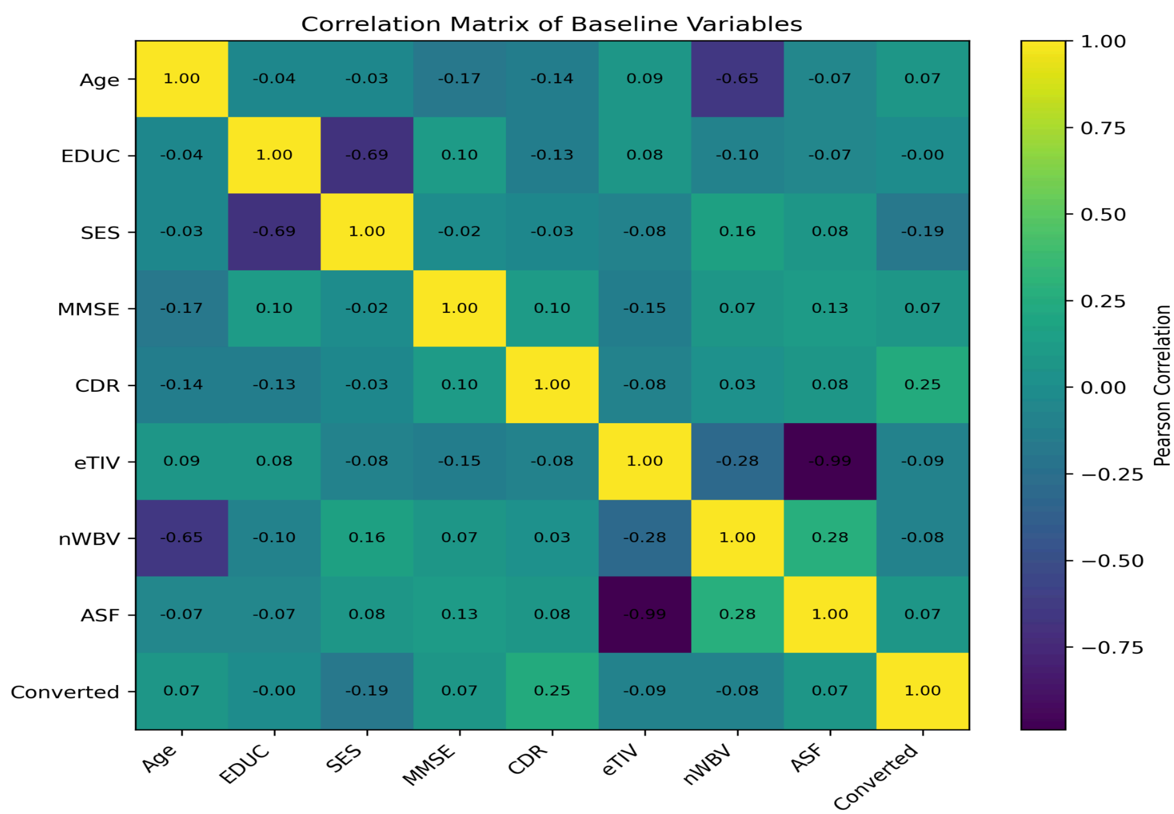
These findings indicate that no individual baseline variable independently distinguished future converters from stable participants within this cohort.

Correlation Analysis

Pearson correlation analysis demonstrated generally weak relationships between individual baseline variables and future dementia conversion. Among the variables examined, Clinical Dementia Rating (CDR) exhibited the strongest positive correlation with future conversion, while socioeconomic status (SES) demonstrated the strongest negative correlation.

Overall, the observed correlation coefficients were modest, indicating that no single demographic, cognitive, or MRI-derived variable provided strong predictive information when considered independently.

Figure 1. Correlation matrix illustrating relationships among baseline demographic, cognitive, MRI-derived, and outcome variables.



Machine Learning Performance

Two supervised machine learning algorithms were evaluated using repeated stratified five-fold cross-validation. Logistic Regression consistently outperformed Random Forest across all evaluated performance metrics.

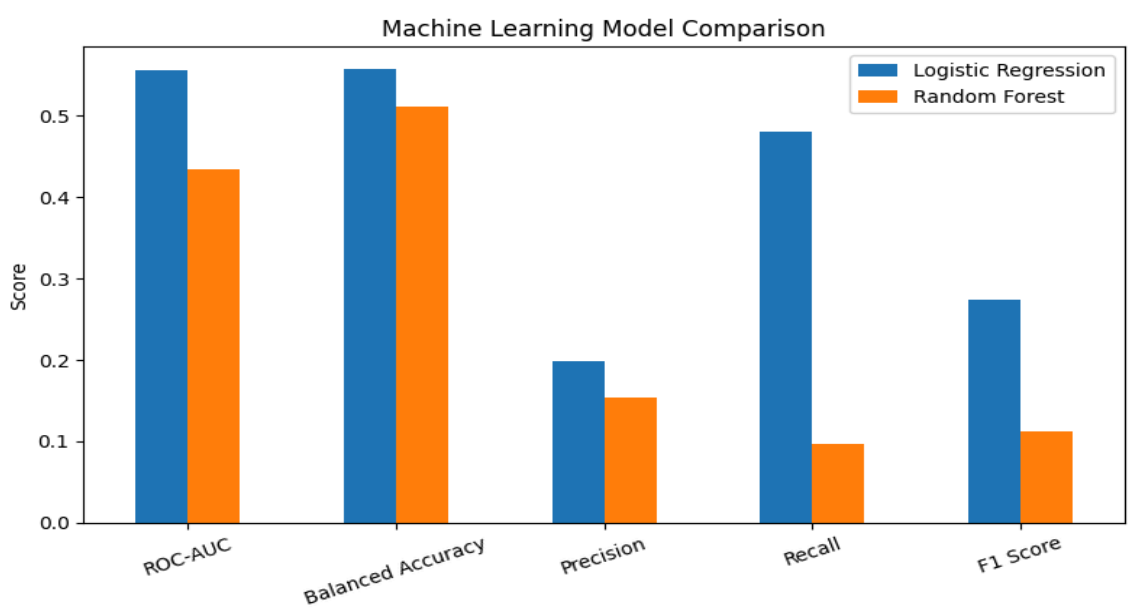
Logistic Regression achieved a mean ROC-AUC of 0.556 ± 0.181, while Random Forest demonstrated lower discrimination with a mean ROC-AUC of 0.474. Similar trends were observed for balanced accuracy, precision, recall, and F1-score.

Although overall predictive performance remained modest, the results suggest that the simpler, more interpretable Logistic Regression model generalized more effectively than the more complex Random Forest model within this relatively small longitudinal cohort.

Table 2. Comparison of cross-validated machine learning performance metrics.

Metric	Logistic Regression	Random Forest
ROC-AUC	0.556	0.474
Balanced Accuracy	0.557	0.502
Precision	0.198	0.122
Recall	0.480	0.060
F1 Score	0.274	0.077

Figure 2. Comparison of Logistic Regression and Random Forest performance across five evaluation metrics. Logistic Regression consistently outperformed Random Forest across all evaluated metrics.



Logistic Regression Interpretation

Examination of the Logistic Regression coefficients demonstrated that baseline Clinical Dementia Rating (CDR), age, and male sex were associated with increased predicted odds of future dementia conversion. In contrast, socioeconomic status, estimated total intracranial volume (eTIV), and years of education were associated with decreased predicted odds.

Permutation feature importance analysis further demonstrated that predictive performance resulted from the combined contribution of multiple variables rather than dependence on a single dominant predictor. These findings support the multifactorial nature of dementia progression and emphasize the importance of integrating demographic, cognitive, and structural MRI information when predicting future disease conversion.

Figure 3. Logistic Regression coefficients for baseline predictors of future dementia conversion. Positive coefficients indicate variables associated with increased predicted odds of future dementia conversion, whereas negative coefficients indicate decreased predicted odds.

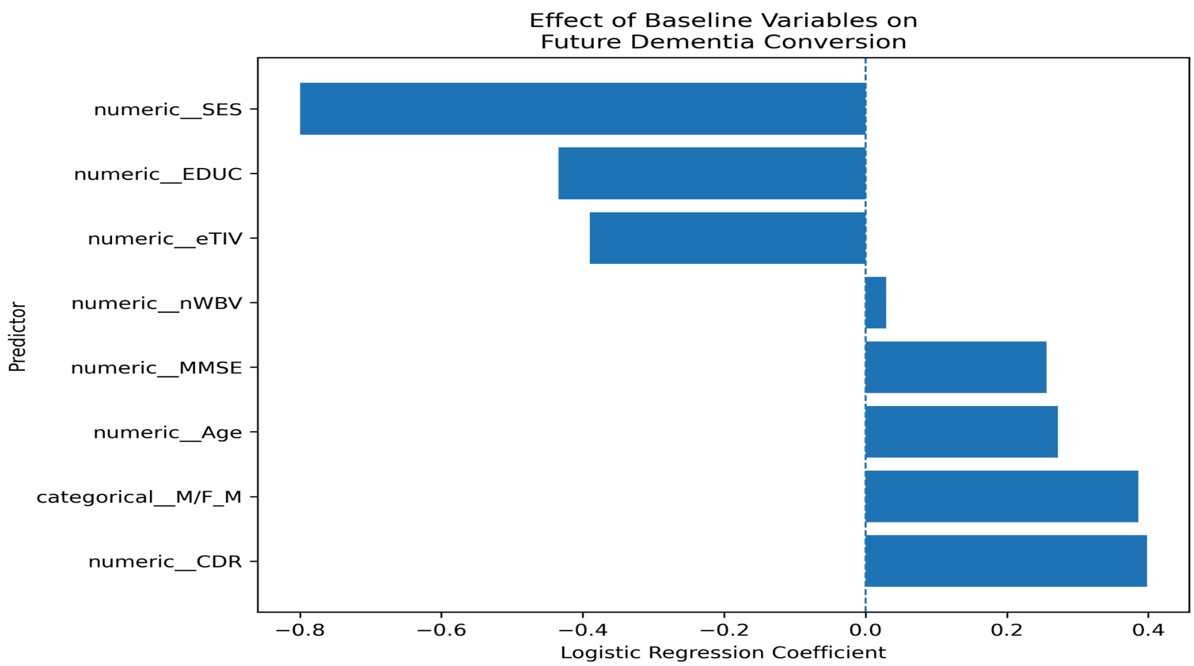
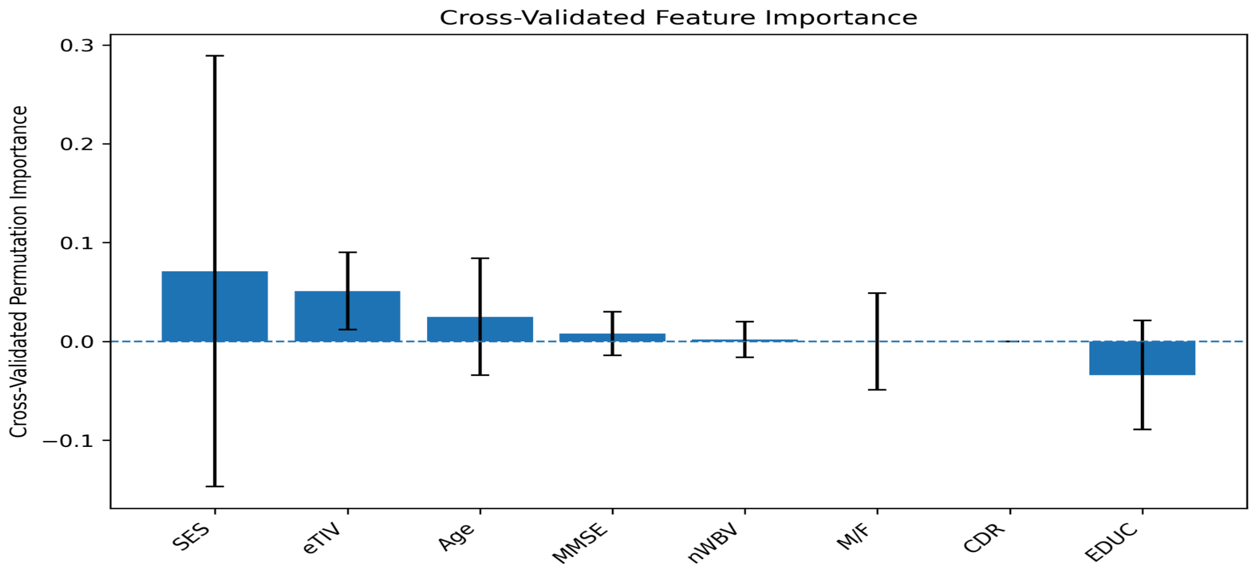


Figure 4. Cross-validated permutation feature importance for the Logistic Regression model. Error bars represent variability across repeated cross-validation folds. Larger positive values indicate greater contribution to predictive performance.



5. Discussion

Overview of Findings

The objective of this study was to investigate whether baseline demographic characteristics, cognitive assessments, and MRI-derived structural biomarkers could predict future dementia conversion using interpretable machine learning. Overall, the findings indicate that baseline clinical and global MRI-derived measurements provided modest predictive information regarding future dementia conversion within the OASIS-2 longitudinal cohort.

Among the evaluated machine learning models, Logistic Regression consistently outperformed Random Forest across all performance metrics despite its comparatively simple model structure. Although predictive performance remained modest, the results demonstrate that interpretable machine learning methods can identify meaningful patterns within longitudinal clinical data while providing insight into the relative contribution of individual predictors.

Interpretation of the Statistical Findings

No individual baseline variable demonstrated a statistically significant difference between stable participants and future converters. This finding is not unexpected given the relatively small number of future converters within the study cohort and the considerable biological variability associated with neurodegenerative disease progression.

Although statistical significance was not achieved, Clinical Dementia Rating (CDR) demonstrated the largest effect size among the evaluated variables. Because CDR reflects subtle functional impairment, this observation is biologically plausible and consistent with the gradual progression of cognitive decline that characterizes dementia.

Similarly, the weak correlations observed between individual predictors and future conversion suggest that dementia progression cannot be adequately explained by a single demographic, cognitive, or MRI-derived measurement. Instead, the disease likely reflects the combined influence of multiple interacting biological processes.

Interpretation of the Machine Learning Results

A key finding of this study was that Logistic Regression consistently outperformed Random Forest despite the latter being a more flexible nonlinear algorithm. This result illustrates an important principle in biomedical machine learning: increasing model complexity does not necessarily improve predictive performance.

The relatively small sample size, particularly the limited number of future converters, likely restricted the ability of Random Forest to learn stable nonlinear relationships. In contrast, Logistic Regression required fewer model parameters and therefore generalized more effectively within the available dataset.

These findings emphasize that model selection should consider both predictive performance and dataset characteristics rather than assuming that more sophisticated algorithms will always produce superior results.

Biological Significance

The observed results support current understanding of dementia as a multifactorial neurodegenerative disorder. Structural brain changes, cognitive decline, demographic characteristics, and functional impairment each contribute to disease progression, but no individual measurement fully captures future dementia risk.

The relatively modest predictive performance observed in this study likely reflects the fact that global MRI-derived variables, such as normalized whole-brain volume and estimated total intracranial volume, provide only indirect measures of the complex biological processes underlying neurodegeneration. More localized structural biomarkers, including hippocampal atrophy and cortical thickness, may provide stronger predictive information.

Nevertheless, the present findings demonstrate that combining multiple baseline measurements using interpretable machine learning provides greater predictive value than evaluating individual variables independently.

Comparison with Previous Research

Previous studies have reported higher predictive performance for dementia conversion by incorporating multimodal biomarkers, including regional MRI measurements, positron emission tomography (PET), cerebrospinal fluid biomarkers, and genetic information (Moradi et al., 2015; Ding et al., 2019). In contrast, the present study intentionally evaluated prediction using readily available demographic characteristics, cognitive assessments, and global MRI-derived measurements.

Although the predictive performance observed here was lower than that reported in some large multicenter studies, the results demonstrate that meaningful predictive information can still be extracted from commonly available baseline variables. Furthermore, the use of an interpretable model provides greater transparency regarding the contribution of individual predictors, an important consideration for potential clinical applications

6. Limitations

Study Limitations

Although this study demonstrates that baseline demographic, cognitive, and MRI-derived variables contain modest predictive information regarding future dementia conversion, several limitations should be considered when interpreting these findings.

First, the final prediction cohort consisted of only 86 participants, including 14 future converters. The limited number of conversion events reduced statistical power and likely contributed to variability in model performance across cross-validation folds. Larger longitudinal cohorts would provide more stable estimates of predictive performance.

Second, this study relied on global MRI-derived structural measurements, including estimated total intracranial volume (eTIV), normalized whole-brain volume (nWBV), and Atlas Scaling Factor (ASF). These variables provide broad measures of brain structure but do not capture regional neurodegenerative changes that may occur during the earliest stages of Alzheimer's disease. Biomarkers such as hippocampal volume, cortical thickness, and ventricular enlargement may offer greater predictive value.

Third, only baseline measurements were included in the prediction models. While this design reflects a realistic clinical prediction scenario, it does not incorporate longitudinal changes in cognition or brain structure that may improve prediction accuracy over time.

Finally, model performance was evaluated using repeated cross-validation within a single dataset. External validation using an independent longitudinal cohort would be necessary before considering clinical implementation.

7. Future Directions

Future Research

The findings of this study suggest several opportunities for future investigation. Incorporating regional neuroimaging biomarkers, particularly hippocampal volume and cortical thickness, may improve prediction of future dementia conversion by capturing structural changes that occur early in disease progression.

Future studies may also benefit from integrating multimodal biomarkers, including amyloid PET imaging, tau PET imaging, cerebrospinal fluid biomarkers, blood-based biomarkers, and genetic risk factors such as the APOE ε4 allele. Combining these complementary sources of information may improve predictive performance while providing a more comprehensive understanding of disease progression.

Another important direction involves modeling longitudinal changes rather than relying exclusively on baseline measurements. Tracking changes in cognitive assessments and MRI-derived biomarkers over time may provide greater sensitivity for identifying individuals at increased risk of dementia conversion.

Finally, future work should evaluate these approaches using larger, multicenter cohorts and external validation datasets to improve model generalizability and better assess potential clinical applications.

8. Conclusion

Conclusions

This study investigated whether baseline demographic characteristics, cognitive assessments, and MRI-derived structural biomarkers could predict future dementia conversion within the OASIS-2 longitudinal dataset using interpretable machine learning methods.

Although no individual baseline variable demonstrated statistically significant differences between stable participants and future converters, the combination of demographic, cognitive, and MRI-derived measurements provided modest predictive information. Logistic Regression consistently outperformed Random Forest across multiple evaluation metrics, suggesting that simpler, interpretable models may generalize more effectively than more complex algorithms when applied to relatively small biomedical datasets.

Beyond model performance, this study demonstrates the value of integrating traditional statistical analyses with interpretable machine learning to investigate clinically relevant neuroscience questions. The results reinforce the complexity of dementia progression and highlight the need for larger longitudinal datasets, multimodal biomarkers, and continued development of transparent predictive models.

Overall, this project illustrates both the potential and the challenges of applying machine learning to neurodegenerative disease research. While additional work is needed before such models could support clinical decision-making, the findings contribute to ongoing efforts to better understand and predict the progression of dementia.

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